

- 1) Water is universal solvent due to-----? **polarity**
- 2) Oxidation of gold is -----? **Non spontaneous process**
- 3) Two systems that are each found to be in thermal equilibrium with a third system will be found to be in thermal equilibrium with each other is law?
Zeroth law
- 4) In calorimetry we study the change in? **heat**
- 5) Freezing point is aProperty? **cogitative**
- 6) The magnitude of force that controls the shape of the liquid is called? **Surface tension**
- 7) The dipole moment is found in? **I guess NaCl or Water**
- 8) The amount needed to increase the temperature of one mole of a substance is called? **Molar heat capacity**
- 9) Gibbs free energy is a? **state function**
- 10) Fugacity Measuresof a Gas. **Non ideality**
- 11) $Pv = nRT$? **Ideal gas equitation.**
- 12) A key property in thermodynamics is-----? **Temperature**
- 13) Two thermodynamics systems in equilibrium have same -----? **Temperature**
- 14) If we bring any change at equilibrium the system compensate for change this is called? **Le Chatelier's Principle**
- 15) The mass/mass percentage is among the-----? **Relative amounts**
- 16) The quantitative study of heat changes is called-----? **Thermochemistry**
- 17) Density and pressure are examples of is -----? **Intensive thermodynamic property**
- 18) The system in which heat change is positive is? **Exothermic**

19) Define physical chemistry?

Physical Chemistry:

Physical chemistry is the study of the underlying physical principles that govern the properties and behavior of chemical systems.

20) What is the effect of the catalyst on the of activation energy?

However, if a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed. Enzymes can be thought of as biological catalysts that lower activation energy.

- A catalyst provides a reaction route with a lower activation energy.
- Suppose that the catalyzed activation energy is to 25 kJ mol⁻¹.
- The calculation is represented at 293 K:

$$e^{-\frac{E_A}{RT}} = e^{-\frac{25000}{8.31 \times 293}}$$
$$= 3.47 \times 10^{-5}$$

- Compared with the corresponding value for activation energy of 50 kJ mol⁻¹, there is a significant increase in the fraction of molecules able to react.

- There are almost 30,000 times more molecules which can react in the presence of the catalyst compared to having no catalyst (using our assumptions about the activation energies).
- These calculations can also be done in reverse: given the rate of reaction or rate constants at several temperatures, the activation energy can be calculated.

21) What is pseudo first order reaction?

Under certain conditions, the second order kinetics can be well approximated (might appear in an experiment) as first order kinetics. This is called pseudo first order reaction. These Pseudo first order reactions greatly simplify quantifying the reaction dynamics. The functional form of the decay kinetics is similar to the first order kinetics and the system is said to operate under pseudo first order kinetics. In a pseudo first order reaction, we can manipulate the initial concentrations of the reactants.

22) Define solubility?

The solubility of a solution is a measure of how much of the solute can be dissolved into the solvent. The solution reaches a point called the saturation point when no more solute will be accepted by the solvent. Any further addition of solute will result in solid solute mixed in with the saturated solution. Each solvent and solute pair has a characteristic solubility at a given temperature. Usually as you increase the temperature, an increased amount of solute will be able to dissolve.

23) What are elementary steps for the reaction mechanism?

Based on numbers of molecules involved in the elementary step, there are three kinds of elementary steps:

1. Unimolecular step
2. Bimolecular step
3. Trimolecular step

1. Unimolecular Step

When a molecule or ion decomposes by itself, such an elementary step is called a unimolecular step (or process). A unimolecular step is always a first order reaction. The following examples are given to illustrate this point:



Or in general

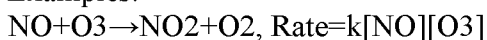


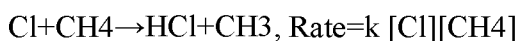
A* represents an excited molecule.

2. Bimolecular Step

A bimolecular process involves two reacting molecules or ions. The rates for these steps are 2nd order, and some examples are given to illustrate how you should give the rate expression. The simulation illustrates a bimolecular process.

Examples:



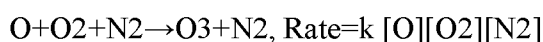


In general:



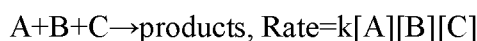
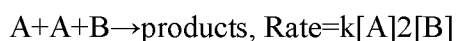
3. Trimolecular Step (process)

A trimolecular process involves the collision of three molecules. For example:



The N_2 molecules in the above trimolecular elementary steps are involved with energy transfer. They cannot be canceled. They are written in the equation to give an expression for the *Rates*.

In general, trimolecular steps may be,



Three molecules colliding at an instant is rare, but occasionally these are some of the ways reactions take place.

24) Write Properties of colloids?

- The particles of dispersant are between about 500 nm to 1 nm in diameter.
- The mixture does not separate on standing in a standard gravity condition. (One 'g'.)
- The mixture does not separate by common fiber filter, but might be filterable by materials with a smaller mesh.
- The mixture is not necessarily completely homogeneous, but usually close to being so.
- The mixture may appear cloudy or almost totally transparent, but if you shine a light beam through it, the pathway of the light is visible from any angle. This scattering of light is called the Tyndall effect.
- There usually is not a definite, sharp saturation point at which no more dispersant can be taken by the dispersing agent.
- The dispersant can be coagulated, or separated by clumping the dispersant particles with heat or an increase in the concentration of ionic particles in solution into the mixture.
- There is usually only small effect of any of the colligative properties due to the dispersant.

25) What is mechanism of stopped flow method of concentrations?

- ☐ ☐ This technique involves two reactants held in separate reservoirs that are prevented from freely flowing by syringe pumps.
- ☐ ☐ The reaction is initiated by depressing the reactant syringes, and thus releasing the reactants into the connecting "mixing chamber" where the solutions are mixed.
- ☐ ☐ The reaction is monitored by observing the change in absorbance of the reaction solution as a function of time.
- ☐ ☐ As the reaction progresses it fills the "stop syringe" which then expands until it hits a block at the point when the reaction has reached a continuous flow rate, thereby stopping the flow and the reaction, and thus allowing the researcher to calculate the exact initial rate of reaction.

□□The Stopped Flow technique works because within milliseconds of combining the two reactants the absorbance can be read.

□□In addition, the stop syringe assures for a steady rate of flow pas the spectrophotometer so that reactants are being added to solution and forming products at a consistent rate.

26) What are Azeotropes?

An azeotrope is a mixture that exhibits the same concentration in the vapor phase and the liquid phase. Azeotropes are a mixture of at least two different liquids. This is in contrast to ideal solutions with one component typically more volatile than the other; this is the reason we use distillation to separate materials. If the mixture forms an azeotrope, the vapor and the liquid concentrations are the same, which preventing separation via this approach. Their mixture can either have a higher boiling point than either of the components or they can have a lower boiling point. Azeotropes occur when fraction of the liquids cannot be altered by distillation. Typically when dealing with mixtures, components can be extracted out of solutions by means of Fractional Distillation, or essentially repeated distillation in stages (hence the idea of 'fractional'). The more volatile component tends to vaporize and is collected separately while the least volatile component remains in the distillation container and ultimately, the result is two pure, separate solutions.

27) What are the limitations of thermodynamics?

Thermodynamics is a powerful approach toward understanding chemical reactions, but only provides part of the picture.

Specifically:

- Thermodynamics only points the way
- Thermodynamics says nothing about how long it takes to get there
- The stoichiometric equation for the reaction says nothing about its mechanism

Thermodynamics only points the way

Chemical change is driven by the tendency of atoms and molecules to rearrange themselves in a way that results in the maximum possible dispersion of thermal energy into the world. The observable quantity that measures this spreading and sharing of energy is the free energy of the system. As a chemical change takes place, the quantities of reactants and products change in a way that leads to a more negative free energy. When the free energy reaches its minimum possible value, there is no more net change and the system is said to be in equilibrium. The beauty of thermodynamics is that it enables us to unfailingly predict the net direction of a reaction and the composition of the equilibrium state even without conducting the experiment; the standard free energies of the reactants and products, which can be independently measured or obtained from tables, are all we need.

The stoichiometric equation for the reaction says nothing about its mechanism:

The term "mechanism" refers to, "who does what to whom". Think of a reaction mechanism as something that goes on in a "black box" that joins reactants to products. □The inner workings of the black box are ordinarily hidden from researchers, are highly unpredictable, and can only be inferred by indirect means. The stoichiometric equation for the reaction says nothing about its mechanism!

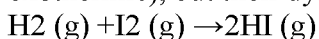
Thermodynamics says nothing about how long it takes to get there:

It is worth noting that the concept of "time" plays no role whatsoever in thermodynamics. But kinetics is all about time. The "speed" of a reaction (how long it takes to reach equilibrium) bears no relation at all to how spontaneous it is (as given by the sign and value

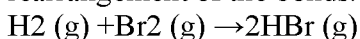
of ΔG°) or whether it is exothermic or endothermic (given by the sign of ΔH°). Moreover, there is no way that reaction rates can be predicted in advance; each reaction must be studied individually.

Examples:

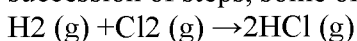
Consider, for example, the gas-phase formation reactions of the hydrogen halides from the elements. The thermodynamics of these reactions are all similar (they are all highly exothermic), but their dynamics (their kinetics and mechanisms) could not be more different.



Careful experiments, carried out over many years, are consistent with the simplest imaginable mechanism: a collision between the two reactant molecules results in a rearrangement of the bonds.



One might be tempted to suppose that this would proceed in a similar way, but experiments reveal that the mechanism of this reaction is far more complex. The reaction takes place in a succession of steps, some of which involve atomic H and Br.



The mechanism of this reaction is different again. Although the first two reactions reach equilibrium in minutes to an hour or so at temperatures of 300 to 600 K, a mixture of hydrogen and chlorine will not react at all in the dark, but if you shine a light on the mixture, it goes off with a bang as the instantaneous reaction releases heat and expands the gas explosively. What is particularly noteworthy is that these striking differences cannot be reliably predicted from theory; they were revealed only by experimentation.

28) Define the half life of reaction, Discuss half life of 0, 1st and 2nd order reactions?**Half life of reaction**

The half life of a reaction, $t_{1/2}$, is the amount of time needed for a reactant concentration to decrease by half compared to its initial concentration.

In zero order reactions/kinetics, the rate of a reaction does not depend on the substrate concentration. In other words, saturating the amount of substrate does not speed up the rate of the reaction. Below is a graph of time (t) vs. concentration [A] in a zero order reaction, several observation can be made: the slope of this plot is a straight line with negative slope equal negative k, the half life of zero order reaction decrease as the concentration decrease. We learn that the zero order kinetic rate law is as followed, where [A] is the current concentration, $[A]_0$ is the initial concentration, and k is the reaction constant and t is time:

$$[A] = [A]_0 - kt$$

We would end up with a formula as such depict how long it takes for the initial concentration to dwindle by half:

$$t_{1/2} = \frac{[A]_0}{2k}$$

The $t_{1/2}$ formula for a zero order reaction suggests the half life depends on the amount of initial concentration and rate constant.

Half Life of 1st Order Reactions

In First order reactions, graph represents the half life is different from zero order reaction in a way that the slope continually decreases as time progresses until it reaches zero. We can also easily see that the length of half life will be constant, independent of concentration. For example, it takes the same amount of time for the concentration to decrease from one point to

another point. In order to solve the half life of first order reactions, we recall that the rate law of a first order reaction was:

$$[A] = [A]_0 e^{-kt}$$

To find the half life we need to isolate t and substitute $[A]$ with $[A]_0/2$, we end up with an equation looking like this:

$$t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k}$$

The formula for $t_{1/2}$ shows that for first order reactions, the half life depends solely on the reaction rate constant, k . We can visually see this on the graph for first order reactions when we note that the amount of time between one half life and the next are the same. Another way to see it is that the half life of a first order reaction is independent of its initial concentration.

Half Life of 2nd order Reactions

Half life of second order reactions shows concentration $[A]$ vs. time (t), which is similar to first order plots in that their slopes decrease to zero with time. However, second order reactions decrease at a much faster rate as the graph shows. We can also note that the length of half life increase while the concentration of substrate constantly decreases, unlike zero and first order reaction.

In order to solve for half life of second order reactions we need to remember that the rate law of a second order reaction is:

$$\frac{1}{A} = \frac{1}{[A]_0} + kt$$

As in zero and first order reactions, we need to isolate T on its own:

$$t_{1/2} = \frac{1}{k[A]_0}$$

This replacement represents half the initial concentration at time, t (depicted as $t_{1/2}$). We then insert the variables into the formula and solve for $t_{1/2}$. The formula for $t_{1/2}$ shows that for second order reactions, the half life only depends on the initial concentration and the rate constant.

29) What are factors that affect the reaction rate explain?

Factors That Affect Reaction Rates: • Although a balanced chemical equation for a reaction describes the quantitative relationships between the amounts of reactants present and the amounts of products that can be formed, it gives us no information about whether or how fast a given reaction will occur.

- This information is obtained by studying the chemical kinetics of a reaction, which depend on various factors: reactant concentrations, temperature, physical states and surface areas of reactants, and solvent and catalyst properties if either are present.
- By studying the kinetics of a reaction, chemists gain insights into how to control reaction conditions to achieve a desired outcome.

Concentration Effects

- ☐ ☐ Two substances cannot possibly react with each other unless their constituent particles (molecules, atoms, or ions) come into contact.
- ☐ ☐ If there is no contact the reaction will be zero order.
- ☐ ☐ Conversely, the more reactant particles that collide per unit time, the more often a reaction between them can occur.

□□Consequently, the reaction rate usually increases as the concentration of the reactants increases.

Temperature Effects

□□Increasing the temperature of a system increases the average kinetic energy of its constituent particles.

□□As the average kinetic energy increases, the particles move faster and collide more frequently per unit time and possess greater energy when they collide. Both of these factors increases the rate of reaction.

□□Hence the reaction rate of virtually all reactions increases with increasing temperature.

□□Conversely, the reaction rate of virtually all reactions decreases with decreasing temperature.

□□For example, refrigeration retards the rate of growth of bacteria in foods by decreasing the reaction rates of biochemical reactions that enable bacteria to reproduce.

Phase and Surface Area Effects:

- When two reactants are in the same fluid phase, their particles collide more frequently than when one or both reactants are solids (or when they are in different fluids that do not mix).
- If the reactants are uniformly dispersed in a single homogeneous solution, then the number of collisions per unit time depends on concentration and temperature, as we have just seen.
- If the reaction is heterogeneous, however, the reactants are in two different phases, and collisions between the reactants can occur only at interfaces between phases.
- The number of collisions between reactants per unit time is substantially reduced relative to the homogeneous case, and, hence, so is the reaction rate.
- The reaction rate of a heterogeneous reaction depends on the surface area of the more condensed phase
- Automobile engines use surface area effects to increase reaction rates.
- Gasoline is injected into each cylinder, where it combusts on ignition by a spark from the spark plug.
- The gasoline is injected in the form of microscopic droplets because in that form it has a much larger surface area and can burn much more rapidly than if it were fed into the cylinder as a stream.
- Similarly, a pile of finely divided flour burns slowly (or not at all), but spraying finely divided flour into a flame produces a vigorous reaction.

Solvent Effects:

The nature of the solvent also reaction rates of the solute particles.

□□For example, a sodium acetate solution reacts with methyl iodide in an exchange reaction to give methyl acetate and sodium iodide.

□□This reaction occurs 10 million times more rapidly in the organic solvent dimethylformamide [(CH₃)₂NCHO] than it does in methanol (CH₃OH).

□□Although both are organic solvents with similar dielectric constants (36.7 for DMF versus 32.6 for methanol), methanol is able to hydrogen bond with acetate ions, whereas DMF cannot.

□□Hydrogen bonding reduced the activity of the oxygen atoms in the acetate ion.

□□Solvent viscosity is also important in determining rates of reactions.

□□In highly viscous solvents, dissolved particles diffuse much more slowly than in less viscous solvents and can collide less frequently per unit time.

□□ Thus the reaction rates of most reactions decrease rapidly with increasing solvent viscosity.

Catalyst Effects

□□ A catalyst is a substance that participates in a chemical reaction and increases the reaction rate without undergoing a net chemical change itself.

□□ Consider, for example, the decomposition of hydrogen peroxide in the presence and absence of different catalysts.

□□ Because most catalysts are highly selective, they often determine the product of a reaction by accelerating only one of several possible reactions that could occur. Most of the bulk chemicals produced in industry are formed with catalyzed reactions.

□□ Recent estimates indicate that about 30% of the gross national product of the United States and other industrialized nations relies either directly or indirectly on the use of catalysts.

30) Difference between open, closed and isolated system? 3 marks

Open system:

Open system is one where transfer of matter between system and surroundings can occur.

Closed system:

Closed system is one where no transfer of matter can occur between system and surroundings.

Isolated system:

Isolated system is one that does not interact in any way with its surroundings.

An isolated system is obviously a closed system, but not every closed system is isolated.

31) Applications of equilibrium constant? 3 marks

Applications of Equilibrium Constants:

1. The magnitude of the equilibrium constant, K , indicates the extent to which a reaction will proceed:

If K is a large number, it means that the equilibrium concentration of the products is large. In this

case, the reaction as written will proceed to the right (resulting in an increase in the concentration of products)

If K is a small number, it means that the equilibrium concentration of the reactants is large.

In

this case, the reaction as written will proceed to the left (resulting in an increase in the concentration of reactants).

Knowing the value of the equilibrium constant, K , will allow us to determine:

The direction a reaction will proceed to achieve equilibrium

The ratios of the concentrations of reactants and products when equilibrium is reached

2. Predicting the Direction of a Reaction

If $Q = K_c$, then the system is already at equilibrium

If $Q > K_c$, then essentially we have too much product and the reaction will proceed to the left (to

reduce the concentration of product and increase the concentration of product)

If $Q < K_c$, then essentially we have too little product and the reaction will proceed to the right (to

produce more product and decrease the concentration of reactant)

. Calculation of the Equilibrium Concentration of a Reactant or Product

Many types of equilibrium problems deal with determining how much of a product (or reactant)

we will have once a reaction reaches equilibrium.

32) Why ideal gas equation failure? Also describe corrections? 5 marks

Combining all 3 laws...

Thus we get the ideal gas equation:

$$pV = nRT$$

for ideal gas it is assumed as

Ideal gas particles occupy negligible volume

Ideal gas has negligible intermolecular interactions

Ideal gas equation shows deviations from

Avogadro's Hypothesis

Boyle's Law

Charles' Law

So it's called

Failures of ideal gas equation

14. Van der Waals Equation for real gases:

As the ideal gas equation deviates from gas laws

So van der Waals in 1873 modified the ideal-gas equation to give the van der Waals equation for

real gases

Van der Waals:

Modified from ideal gas equation

Accounts for:

Non-zero volumes of gas particles (repulsive effect)

Attractive forces between gas particles (attractive effect)

Attractive effect

Pressure = Force per unit area of container exerted by gas molecules

Dependent on:

Frequency of collision

Force of each collision

Both factors affected by attractive forces

Each factor dependent on concentration (n/V)

Hence pressure changed proportional to $(n/V)^2$

Letting a be the constant relating p and (n/V)

Pressure term, p , in ideal gas equation becomes

$$[p + a(n/V)^2]$$

Repulsive effect:

Gas molecules behave like small, impenetrable spheres

Actual volume available for gas smaller than volume of container, V

Reduction in volume proportional to amount of gas, n

Let another constant, b , relate amount of gas, n , to reduction in volume

Volume term in ideal gas equation, V , becomes $(V - nb)$

Combining both derivations...

We get the Van der Waals Equation

Van der Waal's corrections:

Sir Johannes Diderik van Der Waals in 1873, modified the Ideal

Gas Equation

He proposed the ideal gas equation with

"Van der Waal's corrections"

By adding the necessary Constants Critical Constants



$$\left[p + a \left(\frac{n}{V} \right)^2 \right] [V - nb] = nRT$$

OR

$$\left[p + \frac{a}{V_m^2} \right] [V_m - b] = RT$$

For intermolecular attractions (weak forces)

B. The volume occupied by the gas particles (themselves)

At higher pressure, the volume is much reduced and

at this state the volume of gas molecules Sir Johannes Diderik van Der Waals in 1873, modified the Ideal Gas Equation He proposed the ideal gas equation with

"Van der Waal's corrections"

By adding the necessary Constants

Critical Constants

For intermolecular attractions (weak forces)

B. The volume occupied by the gas particles (themselves)

At higher pressure, the volume is much reduced and

at this state the volume of gas molecules

Becomes "no more negligible" in comparison with the total volume V occupied by the gas.

The "no more negligible" volume is generally denoted by "b" called effective volume.

Therefore the total (actual) volume available in which the molecules are free to move will be

= Total volume (V) — Effective volume (b).

i.e., Correct Volume (V - b)